WORKED SOLUTION

# STEP II 2015, Question 5

Pure • Arctan Series

**STEP II 2015, Question 5**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s2-q5

## The key idea

The tangent addition formula turns a sum of arctangents into a single fraction, so the induction for $\tan S_n$ is a matter of careful algebra. The subtle point is the step from $\tan S_n$ to $S_n$ itself: knowing a tangent only fixes an angle up to multiples of $\pi$, so we must pin down which interval $S_n$ lies in. In part (ii), the angles $\alpha_n$ turn out to be exactly the arctangents of part (i), and the infinite sum is the limit of $S_n$.

## Solution

### Part (i)

Write $\theta_m = \arctan(\frac{1}{2m^2})$, so that $S_n = \theta_1 + \cdots + \theta_n$ and $\tan\theta_m = \frac{1}{2m^2}$.

**The formula for $\tan S_n$.** For $n = 1$, $\tan S_1 = \frac{1}{2} = \frac{1}{1+1}$, so the result holds.

Suppose $\tan S_n = \frac{n}{n+1}$ for some $n \geq 1$. Since $S_{n+1} = S_n + \theta_{n+1}$, the addition formula gives

$$\tan S_{n+1} = \frac{\tan S_n + \tan\theta_{n+1}}{1 - \tan S_n \tan\theta_{n+1}} = \frac{\dfrac{n}{n+1} + \dfrac{1}{2(n+1)^2}}{1 - \dfrac{n}{2(n+1)^3}}.$$

The denominator is not zero, because the product $\frac{n}{n+1} \cdot \frac{1}{2(n+1)^2}$ is less than 1. Multiplying top and bottom by $2(n+1)^3$,

$$\tan S_{n+1} = \frac{(n+1)(2n(n+1)+1)}{2(n+1)^3 - n} = \frac{(n+1)(2n^2+2n+1)}{2n^3+6n^2+5n+2}.$$

**Ask yourself** How do we know the denominator will factorise nicely?

We are aiming for $\frac{n+1}{n+2}$, so the denominator ought to be $(n+2)(2n^2+2n+1)$. Expanding confirms it: $(n+2)(2n^2+2n+1) = 2n^3+6n^2+5n+2$. Cancelling the positive factor $2n^2+2n+1$,

$$\tan S_{n+1} = \frac{n+1}{n+2},$$

which is the result for $n+1$. By induction, $\tan S_n = \frac{n}{n+1}$ for all $n \geq 1$.

**The formula for $S_n$.** From $\tan S_n = \frac{n}{n+1}$ alone we can only say $S_n = \arctan\frac{n}{n+1} + k\pi$ for some integer $k$. To show $k = 0$ we prove, again by induction, that

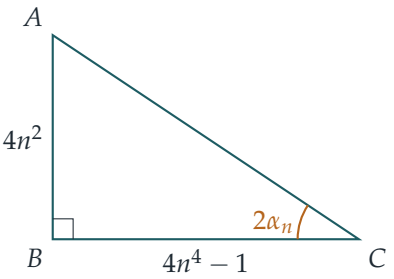
$$S_n = \arctan\frac{n}{n+1}.$$

For $n = 1$, $S_1 = \arctan\frac{1}{2}$, as required. Suppose the result holds for some $n$. Since $0 < \frac{n}{n+1} < 1$, we have $0 < S_n < \arctan 1 = \frac{1}{4}\pi$. Also $0 < \frac{1}{2(n+1)^2} < 1$, so $0 < \theta_{n+1} < \frac{1}{4}\pi$. Adding,

$$0 < S_{n+1} = S_n + \theta_{n+1} < \tfrac{1}{2}\pi.$$

So $S_{n+1}$ lies in $[0, \frac{1}{2}\pi)$, which is exactly the range of arctan on non-negative numbers, and $\tan S_{n+1} = \frac{n+1}{n+2}$ by the first result. As tan is one-to-one on this interval, $S_{n+1} = \arctan\frac{n+1}{n+2}$. This completes the induction.

**Part (ii)**




Since the angle at $B$ is a right angle, the side opposite angle $C$ is $AB$ and the adjacent side is $BC$ (note $4n^4 - 1 > 0$ for $n \geq 1$). So

$$\tan 2\alpha_n = \frac{AB}{BC} = \frac{4n^2}{4n^4-1}.$$

Part (i) suggests trying $\tan\alpha_n = \frac{1}{2n^2}$. With $t = \frac{1}{2n^2}$, the double-angle formula gives

$$\frac{2t}{1-t^2} = \frac{\frac{1}{n^2}}{1-\frac{1}{4n^4}} = \frac{4n^2}{4n^4-1},$$

which matches. To be sure this is the right value and not merely a value, write $t = \tan\alpha_n$. The equation $\frac{2t}{1-t^2} = \frac{4n^2}{4n^4-1}$ rearranges to the quadratic

$$4n^2t^2 + 2(4n^4-1)t - 4n^2 = 0,$$

whose roots have product $-1$. One root is $\frac{1}{2n^2}$, so the other is $-2n^2 < 0$. But $2\alpha_n$ is an angle of a right-angled triangle other than the right angle, so $0 < 2\alpha_n < \frac{1}{2}\pi$, giving $0 < \alpha_n < \frac{1}{4}\pi$ and $t > 0$. Hence $\tan\alpha_n = \frac{1}{2n^2}$ and, as $\alpha_n$ lies in the range of arctan,

$$\alpha_n = \arctan\frac{1}{2n^2}.$$

**Ask yourself** What does an infinite sum mean here, and how does part (i) help?

The infinite sum is the limit of its partial sums, and by part (i) these are

$$\sum_{n=1}^{N}\alpha_n = S_N = \arctan\frac{N}{N+1}.$$

As $N \to \infty$, $\frac{N}{N+1} \to 1$, and since arctan is continuous, $\arctan\frac{N}{N+1} \to \arctan 1 = \frac{1}{4}\pi$. Therefore

$$\sum_{n=1}^{\infty}\alpha_n = \tfrac{1}{4}\pi,$$

as required.

## Checks

- For $n = 1$, the triangle in part (ii) is the familiar 3–4–5 triangle, with $\tan 2\alpha_1 = \frac{4}{3}$. Then $\tan\alpha_1 = \frac{1}{2}$ satisfies $\frac{2\cdot\frac{1}{2}}{1-\frac{1}{4}} = \frac{4}{3}$.
- $S_2 = \arctan\frac{1}{2} + \arctan\frac{1}{8}$, and $\frac{\frac{1}{2}+\frac{1}{8}}{1-\frac{1}{16}} = \frac{10}{15} = \frac{2}{3}$, as the formula predicts.
- Numerically, adding the first 2000 terms gives $S_{2000} \approx 0.785148$, against $\frac{1}{4}\pi \approx 0.785398$. The gap is about $\frac{1}{4000}$, as it should be: the subtraction formula gives $\frac{1}{4}\pi - \arctan\frac{N}{N+1} = \arctan\frac{1}{2N+1}$ exactly.

## Common slips

- Concluding $S_n = \arctan\frac{n}{n+1}$ straight from $\tan S_n = \frac{n}{n+1}$, without showing that $S_n$ lies in $[0, \frac{1}{2}\pi)$.
- Taking $\tan\alpha_n = \frac{1}{2n^2}$ from a lucky guess without ruling out the other root of the quadratic.
- Treating the infinite sum as $S_\infty$ without saying it is the limit of $S_N$.

## Beyond STEP

This is a telescoping series in disguise: since $\frac{1}{2m^2} = \frac{(2m+1)-(2m-1)}{1+(2m+1)(2m-1)}$, each term is $\arctan(2m+1) - \arctan(2m-1)$, and the sum collapses to $\arctan(2n+1) - \frac{1}{4}\pi$, which

equals $\arctan \frac{n}{n+1}$. Defining an infinite sum as a limit of partial sums, and knowing when limits can be passed through continuous functions, is the starting point of a first course in analysis. A good place to read about it is Abbott, *Understanding Analysis*.

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